

# Diving into the Toplayer

Where `<dialog>` s, `[popover]` s, and Modals Live

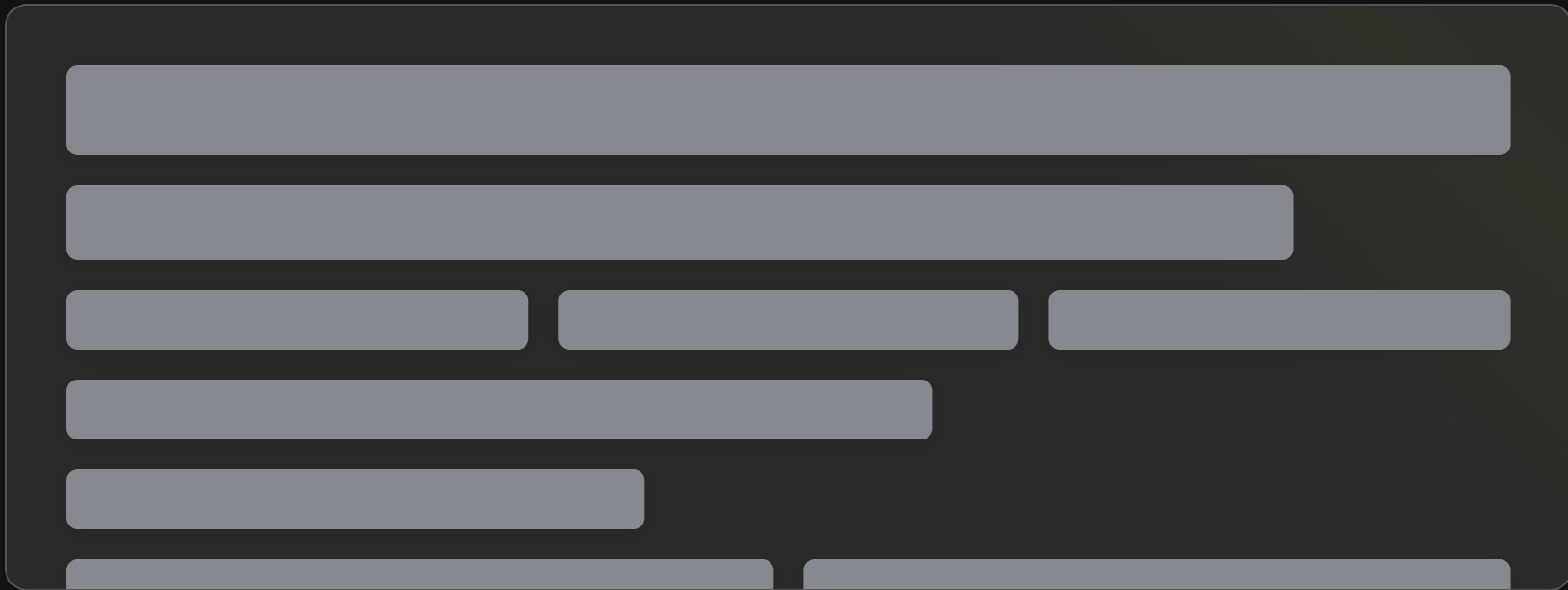
UtahJS Conference

September 12, 2025

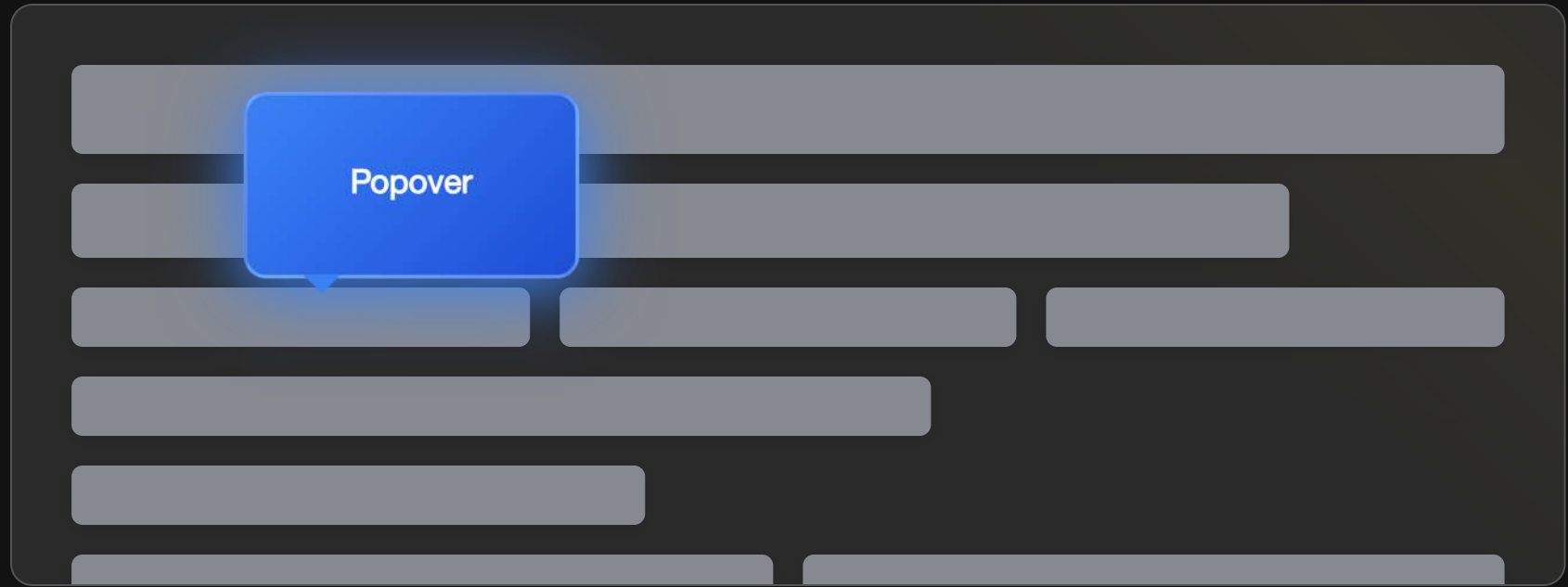
Tim Damen

```
<dialog>
  /* I live in the toplayer 🙋 */
</dialog>
```

# At first the web was flat



# Today, the web can overlap



Andrej Karpathy  
@karpathy · [Follow](#)



Browsing the web, 2021

Watch on X

6:47 AM · Sep 9, 2021



13.4K



Reply



Copy link

[Read 294 replies](#)

## Cookie banners ruined the Internet - Why Europe is losing in tech | DHH and Lex Fridman



# Tim Damen



Front-end Chapter Lead @ ABN ARMO



Accessibility Lead & Advocate



Podcaster @ focustrap



Husband and proud father of a daughter



Passionate about sports and outdoor adventures













We'll focus on the *What*, *Why* and *How* of  
content that overlaps content

With the `<dialog>` element and the `[popover]` attribute

# <dialog>

HTML element - a native dialog element that can be used to create a modal dialog or a non-modal dialog.

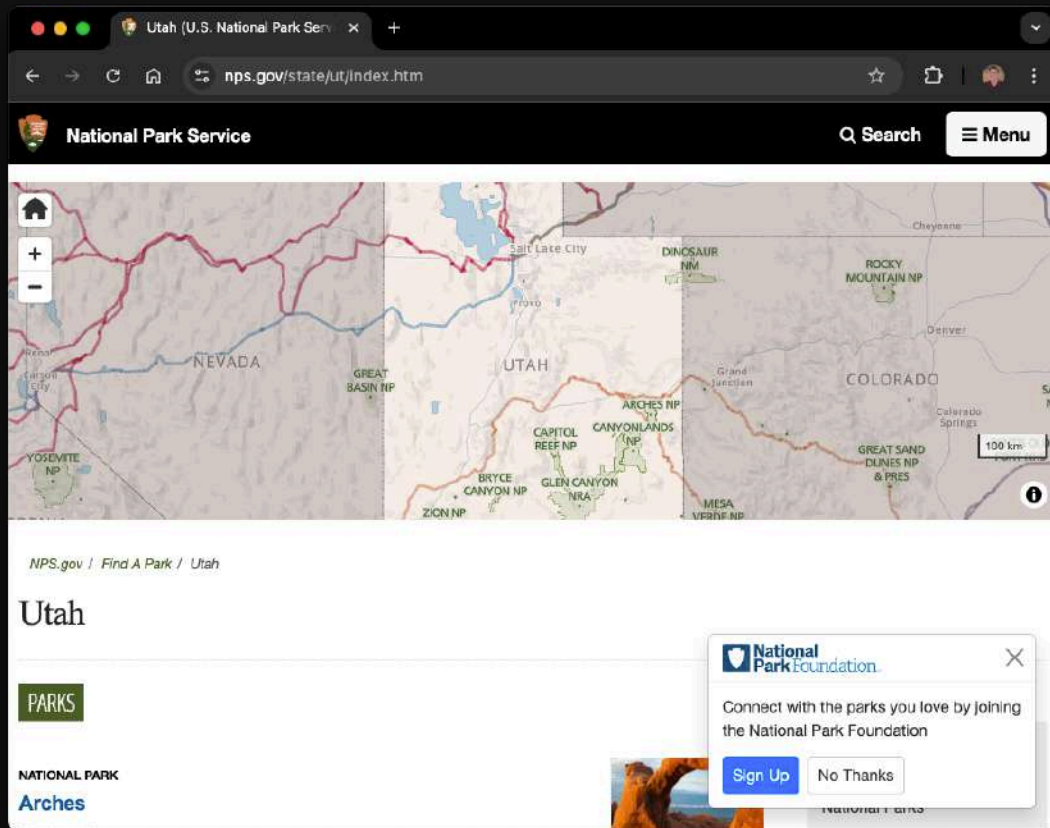
- Supported by all major browsers
  - 2014: Chrome v37
  - 2022: Firefox 98
  - 2022: Safari 15.4
- Good accessibility support
  - 2014-2021: "The a11y community says nope"
  - 2022: "Universal browser support... but still wait"
  - 2023: "Use the dialog element (reasonably) by Scott ohara"

# [popover]

HTML attribute - API in HTML that allows you to create popovers, tooltips, and other overlay content.

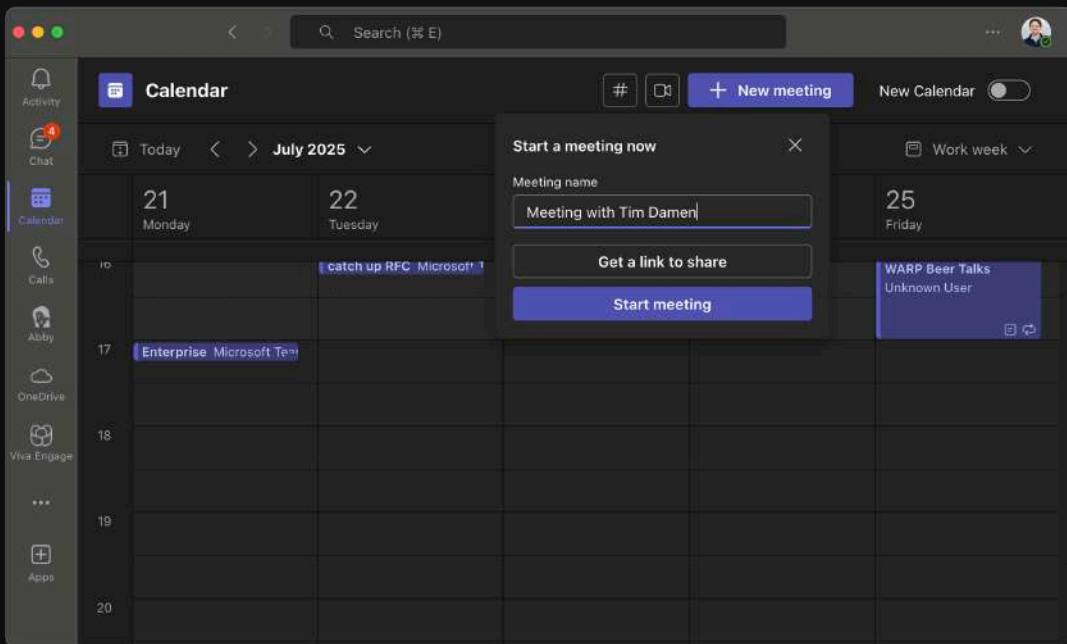
- Supported by all major browsers
  - May 2023: Chrome v114
  - Sep 2023: Safari v17
  - Apr 2024: Firefox v125
- Accessibility was baked in from the start
- Proposed by the Open UI group

# Non-modal dialog

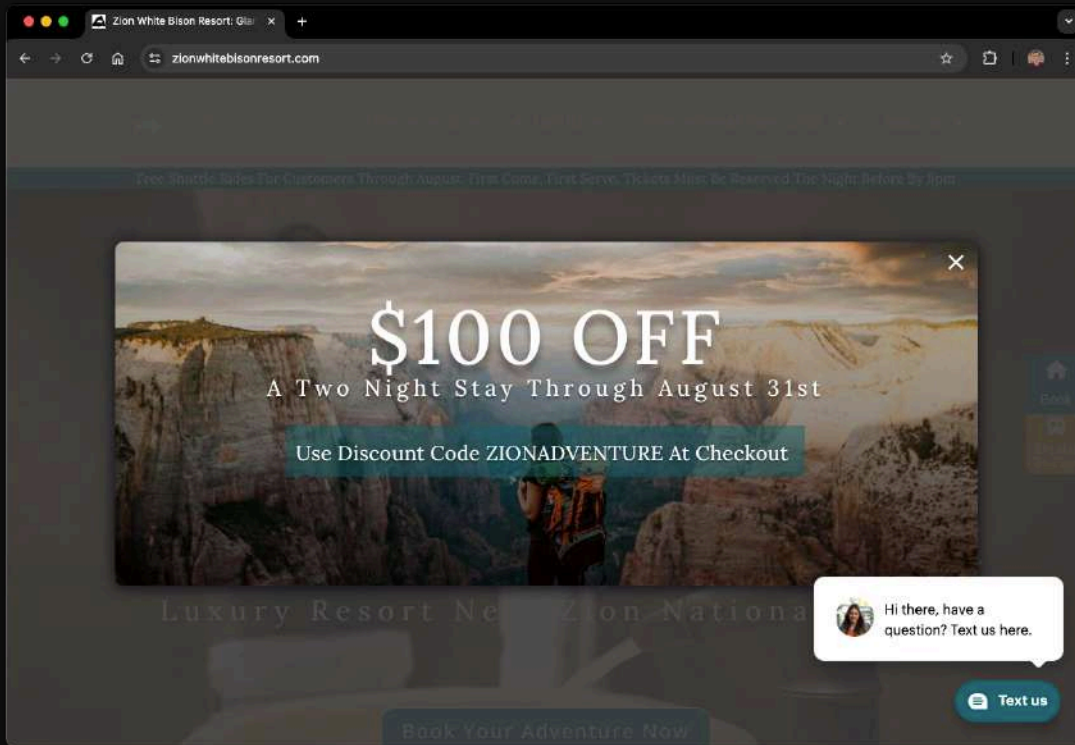




# Popover



# Modal + Popover



# With `z-index` you can stack elements

```
z-index: auto;
```

```
z-index: 1;
```

```
z-index: 3;
```

```
z-index: 5;
```

```
z-index: 7;
```

```
z-index: 6
```

```
z-index: 4
```

```
z-index: 2
```

Change my z-index

```
z-index: auto
```

```
z-index: auto
```

```
z-index: auto
```

# What is the maximum `z-index` value?

```
z-index: 2147483647; /* 2^31 - 1 */
```

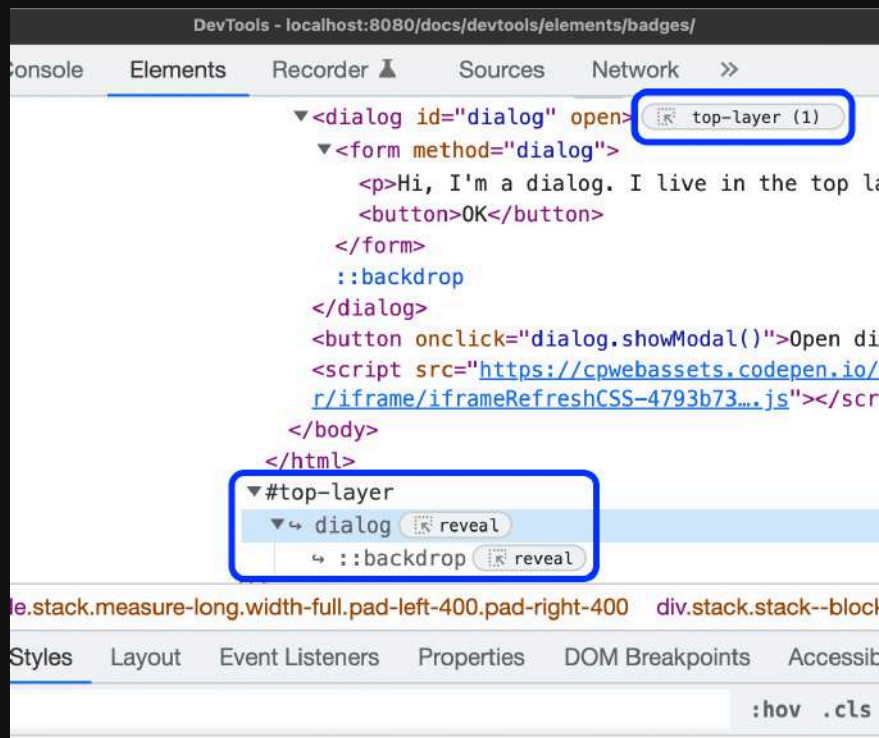


Speak the Number

2,147,483,647

2147483647, the largest number that can be stored in a 32-bit signed integer 🧐

The `#top-layer` is above everything else, it has its own layer above the main document





## #top-layer

On top of z-index stacking

<dialog>  
dialog

requestFullscreen()  
fullscreen

[popover]  
popover

## z-index

z-index: 1 → 2147483647

Dropdown  
z: 10

Overlay  
z: 1000

Tooltip  
z: 100

Notification  
z: 9999

## document

z-index: auto

Content  
main

Header  
header

Navigation  
nav

Sidebar  
aside

Footer  
footer

<dialog> vs [popover]

What a `<dialog>` is

An additional window that (temporarily) overlaps the main content of a web page

# Dialogs

Typically contain:

- Information
- Task
- Action

# <dialog>

## show() vs showModal()

```
// Opens the dialog as a non-modal  
Element.show();
```

```
// Opens the dialog as a modal  
Element.showModal();
```

```
<!-- dialog element -->
<dialog>
  ...
</dialog>
```

- `role="dialog"` is added automatically
- Option to show as modal or non-modal
- `Esc` to close when opened as modal
- Focus is trapped when opened as modal

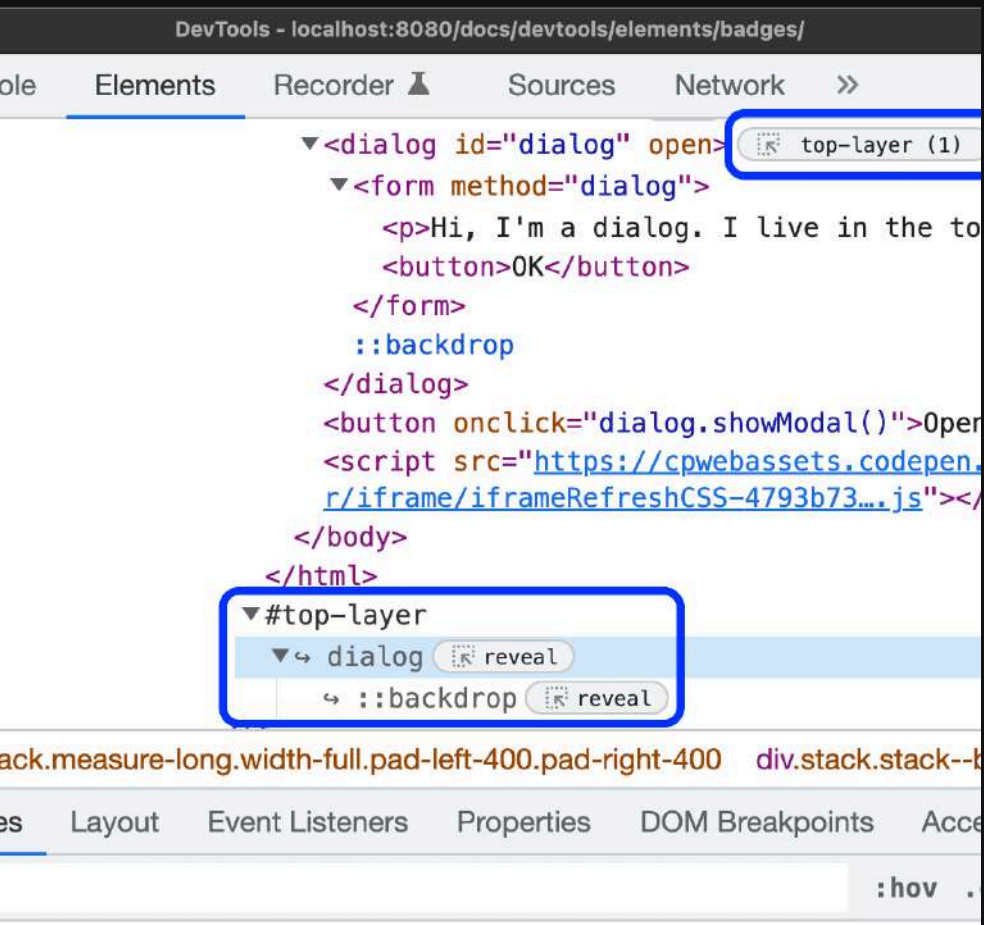


# modal vs non-modal

## HTML Dialog Element Demo

Survey dialog

Session extend



# backdrop

Some component have a backdrop.

Toplayer elements have a built-in styleable pseudo-element

`::backdrop`

`<dialog>` vs `[popover]`

What a [popover] is

A floating overlay that provides information or actions when triggered, appearing above the main content without blocking the user's workflow.

"a popover 'pops'" - Tim Damen

# [popover]

HTML attribute that can be applied to any element, it adds a set of behaviors to the element

```
<div popover>...</div>
```

```
<dialog popover>...</dialog>
```

Popover does not add a `role`. It adopts the role of the element it is applied to.

Depending on the popover type (default is `popover="auto"`), it will add:

- light dismiss
- keyboard behavior (`Esc` to close)
- added to toplayer
- (Usually) Closes other popovers when opened

# [popover] implementation

Requires no JavaScript, just HTML

```
<button>
  Toggle popover
</button>

<div>
  ...
</div>
```

# [popover] implementation

Requires no JavaScript, just HTML

```
<button popovertarget="x" popovertargetaction="show">  
  Open popover  
</button>  
  
<div popover id="x">  
  ...  
</div>
```

Open popover

# [popover] implementation

Requires no JavaScript, just HTML

```
<button popovertarget="x" popovertargetaction="show">
  Open popover
</button>

<!-- Or open with Javascript -->
<button onclick="document.getElementById('x').showPopover()">
  element.showPopover();
</button>

<div popover id="x">
  ...
</div>
```

Open popover

element.showPopover();



localhost:3031/32

Relaunch to update

Elements

31 24 57

⚙️ ⋮ ✕

style="color: black; padding: 2px; background-color: rgb(240, 240, 240);">⌵

element.showPopover(); </button>

▶ <div data-v-e16fe63a popover id="x">⋮

</div> == \$0

dev-page.slidev-page-32 div.slidev-layout.default div#x

Styles

Computed

Layout

Event Listeners

>>

Filter

:hov .cls + 🖨 📄

border-color: ▶

■ var(--un-default-border-color, ■ #e5e7eb);

}

[popover]:popover-open:not(dialog) user agent stylesheet

{

overlay: auto !important;

}

[popover] { user agent stylesheet

position: fixed;

width: fit-content;

height: fit-content;

color: canvastext;

background-color: canvas;

inset: ▶ 0px;

margin: ▶ auto;

border-width: ▶ initial;

border-style: ▶ solid;

border-color: ▶ initial;

border-image: ▶ initial;

padding: ▶ 0.25em;

overflow: ▶ auto;

}

div { user agent stylesheet

[popover] implementation

Requires no JavaScript, just HTML

<button popovertarget="x" popovertargetaction="show">

Open popover

</button>

<!-- 01

Lorem Ipsum is simply dummy text of the printing and typesetting industry.

<button onclick="document.getElementById('x').showPopover()">

element.showPopover();

</button>

<div popover id="x">

...

</div>

Open popover

element.showPopover();

div#x 596.08 x 35.2

32 / 74

# [popover] still has some positioning flaws

**But there is hope! You have got two options:**

- Use a JS library for positioning like `floating-ui`
- Use `CSS anchor()`

`CSS anchor()` is not supported by all browsers yet 😞

```
<button popovertarget="my-tooltip">
  <p>?</p>
</button>

<div id="my-tooltip" class="tooltip" popover>
  <p>I am a tooltip with more information. I am a
  tooltip with more information. I am a tooltip with
  more information. I am a tooltip with more
  information.<p>
</div>
```



# <dialog> vs [popover] vs modals

| <code>&lt;dialog&gt; s</code><br><code>&lt;dialog&gt; with show()</code> | <code>[popover] s</code> | modals<br><code>&lt;dialog&gt; with showModal()</code> |
|--------------------------------------------------------------------------|--------------------------|--------------------------------------------------------|
| Explicit dismiss                                                         | Light dismiss            | Explicit dismiss                                       |
| z-index                                                                  | #top-layer               | #top-layer                                             |
| ✗                                                                        | ::backdrop ?!            | ::backdrop                                             |
| Keyboard assistance                                                      | Keyboard assistance      | Keyboard assistance & focus trap                       |

# The Evolution of Layers on the Web

From z-index clashes to native browser solutions

NEW

1998

## The birth of z-index

The z-index CSS property was introduced in 1998 as part of the CSS Level 2 (CSS2) specification. It was part of the original positioned layout system that included position: absolute, position: relative, and the concept of stacking contexts.

```
z-index: 1;  
position: absolute;  
position: relative;
```

# Thank you! 🙏

Let's stay in touch!

